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Working through the different preprocessing techniques gave me a much clearer sense of how important it is to choose the right level of cleaning for the right task. I used to think preprocessing was mostly about "cleaning up the text," but now I see that every choice—stemming vs. lemmatization, minimal vs. aggressive cleaning, keeping vs. removing emojis—has real consequences for how a model interprets meaning.

One of the biggest insights I gained was that **there's no single "best" preprocessing pipeline**. Instead, the right approach depends entirely on the goal. If I need nuance, tone, or emotional signals, then minimal or standard cleaning makes sense. If I'm building something like a search engine or a topic model, then aggressive cleaning can actually help by reducing noise and shrinking the vocabulary. Seeing how drastically the token counts changed across text types really drove this home.

I also learned how easy it is to accidentally remove information that matters. Emojis, hashtags, and punctuation aren't just decoration—they carry sentiment, emphasis, and context. When I saw how much meaning disappeared after aggressive cleaning, it made me rethink the assumption that "cleaner is always better." Sometimes the "messy" parts of language are exactly what the model needs.

There were definitely challenges along the way. Getting the preprocessing functions to behave consistently across different text types took some trial and error. I also ran into issues where removing too much caused the output to feel empty or misleading. Debugging those moments helped me understand the pipeline more deeply. And comparing stemming to lemmatization made me appreciate how crude stemming can be—those weird fragments like *"fli"* really highlight why meaning-preserving techniques matter.

Finally, I found a lot of connections to real-world applications. Customer reviews, social media posts, academic writing, and news articles all behave differently under preprocessing. It made me realize that NLP systems in the real world must be tailored to their domain. A chatbot needs nuance. A search engine needs speed. A sentiment classifier needs emotional cues. There's no one-size-fits-all solution.

Overall, this project helped me learn and reinforce previous learnings of preprocessing and also did not spook me when an error popped up. When I don't feel panic when an error pops up, I know that I gained some experience and have more tools to address the error. One time, it was just needing to reconnect and reload. Simple.